

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Fourth Semester of B.Tech. Theory (IT) Examination****May 2015****MA206.01 Statistical & Numerical Methods****Date: 13/05/2015, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks:70****Instructions:**

1. This question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever if required.
4. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Write whether following statements are true or false. [07]

- (a) If the mean of 5 values of variable x is 8, and $\sum_{i=1}^4 x_i = 20$, then $x_5 = 20$.
- (b) The number of ways 52 cards can be divided amongst four players so that each may have 13 cards is equal to $\frac{13! \times 13! \times 13! \times 13!}{52!}$
- (c) If A and B are two events on sample space S with probability function $P(\cdot)$, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- (d) Let the least squares lines be $y = 3 + 1.2x$ and $x = 7 + 0.3y$, then $r_{xy} = 0.6$
- (e) The standard deviation of 6 values of variable is 10, if all values of variable are multiplied by -2, the new standard deviation of changed values of variable is -20.
- (f) The geometric mean of 4 values of variable x is 5, if all values of variable are multiplied by 2, the new geometric mean of changed values of variable is 10.
- (g) The value of rank correlation coefficient between n pairs of ranks

$$(x_i, y_i), i = 1, 2, \dots, n \text{ is given by } r_R = 1 - \frac{6 \sum_{i=1}^n (x_i - y_i)^2}{n(n^2 - 1)}.$$

Q.2. Answer the following questions. [14]

- (a) A computer program is tested by 2 independent tests. When there is an error, these tests will discover it with probabilities 0.4, and 0.5, respectively. Suppose that the program contains an error. What is the probability that it will be found by, [03]
 - (i) at least one test?
 - (ii) exactly one test?

- (b) An investigation on CPU performance gives the data on minimum channels in units of different models of CPU: [04]

Table 1: Minimum number of channels

1	1	5	12	3	1	12	3	2	1	1	4	1	12	3	1	1	6	3	8
1	1	5	12	3	1	1	3	3	1	3	4	1	1	1	1	2	1	4	8
8	1	6	12	3	1	1	3	2	1	3	7	1	1	1	1	2	1	1	8
1	1	6	12	3	1	1	6	1	6	3	5	1	1	1	1	0	6	1	1
1	1	1	12	3	1	1	6	3	12	3	8	1	3	1	1	0	1	1	1

Construct the Frequency distribution of number of minimum channels in units from table 1

- (c) The problem consists in classifying all the blocks of the page layout of a document that has been detected by a segmentation process. This is an essential step in document analysis in order to separate text from graphic areas. The five classes are: text (1), horizontal line (2), picture (3), vertical line (4) and graphic (5). The following data show Percentage of black pixels within the block (blackpix / area) [07]

Table 2: Blocks Classifications:Percent of black pixels within block

0.400	0.429	0.287	0.371	0.500	0.550	0.417	0.333	0.400	0.486
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Determine following summary measures on number of percent black pixel based on table 2

- (i) Mean (ii) Median (iii) Standard deviation

OR

Q.2. Answer the following questions. [14]

- (a) Consider the **if** statement in a program [03]

if B **then** s_1 **else** s_2

The random experiment consists of “observing” two successive execution of the statement.

- (i) How many possible outcomes of this experiment? Write Sample Space S of outcomes of this experiment.
(ii) Suppose E_1 described by the statement “At least one execution of the statement s_1 ” and E_2 described by the statement “Statement s_2 is ex-

ecuted the first time ", then compute $P(E_1)$ and $P(E_2)$ assuming all outcomes are equally likely.

- (b) Write important properties of Karl Pearson Correlation coefficient. [04]
 (c) The failure rate of a certain electronic device is suspected to increase linearly with its temperature. [07]

Table 3: Failure rate versus temperature

Temp° F	55	65	75	85	95	105
Failure rate 10^6	1.92	1.94	1.97	2.01	2.015	2.025

- (i) Construct the scatter plot of the data.
 (ii) Determine a least-squares linear line through the data in table 3

Q.3. Answer the following questions. [14]

- (a) Assume that the number of messages input to a communication channel in an interval of duration t seconds is Poisson distributed with parameter $0.3t$. Compute the probabilities of the following events: [05]
 (i) Exactly two messages will arrive during 8 second interval.
 (ii) At most 2 messages arrive in a period of 12 second.
- (b) Consider a time-sharing computer system that allocates a finite quantum (or time slice) to a job scheduled for processor service, in an attempt to give fast service to requests for trivial processing. Observe $n = 6$ time-slice terminations, with success = "job has completed processing" and failure = "job still requires processing and joins the tail end of the ready queue of processes." Let p be the probability that "job has completed processing". If $p = 0.4$ then Write the formula to compute $P(X = r)$, r jobs have completed processing and compute the probability that [05]
 (i) no job has completer processing
 (ii) Exactly three jobs have completed processing
 (iii) atleast 2 jobs have completed processing
- (c) Assume that lifetime of IC chips manufactured by a semiconductor manufacture are approximately Normally distributed with $\mu = 5 \times 10^6$ hours and $\sigma = 5 \times 10^5$ hours. A mainframe manufacturer requires that at least 95% of batch should have lifetime grater than 4×10^6 hours. Will the deal be made? You are given $P(Z \leq 2) = 0.97725$ [04]

OR

Q.3. Answer the following questions. [14]

- (a) Define Uniform distribution over the interval (a, b) , $a < b$. Determine mean of Uniform distribution. If a random variable X is uniformly distributed over $(0, 10)$, calculate the probability $P\{3 \leq X \leq 8\}$ [07]
- (b) Define Exponential distribution. Determine the mean of exponential distribution. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 0.8$. What is the probability that a repair time does not exceed 2 hours? [07]

SECTION II

Q.4. Write whether following statements are true or false. [07]

- (a) If $f(x)$ is real and continuous in the interval (a, b) and $f(a)$ and $f(b)$ have opposite signs, then there is atleast one real root between a and b
- (b) The Newton Raphson formula to find real root of algebraic equation is $x_{i+1} = x_i - \frac{f'(x_i)}{f(x_i)}$, $i = 1, 2, \dots$
- (c) If X is true value of quantity and $\hat{X}$ is its approximation, then relative error is defined as $X - \hat{X}$.
- (d) The relationship between shift operator E , Δ is $E \equiv \Delta - 1$.
- (e) With usual notations one can write $\Delta y_0 = \nabla y_1$
- (f) Assume we have a table of values (x_i, y_i) , $i = 0, 1, 2$ of a function $y = f(x)$. The coefficient of y_1 in Lagrange interpolation formula is $\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$
- (g) Assume we have a table of values (x_i, y_i) , $i = 0, 1, 2, 3$ of a function $y = f(x)$ and the values of x are equally spaced. From the table values, we can obtain polynomial of degree atmost 3 using Newton's forward difference interpolation.

Q.5. Answer the following questions. [14]

- (a) Given the table of values [07]

x	2	2.5	3.2
$y = \ln x$	0.69315	0.91629	1.16315

evaluate $\ln 2.7$ using Lagrange's interpolation formula.

- (b) Find a root of the equation $x \sin x + \cos x = 0$ using Newton Raphson method. Perform four iterations starting with $x_0 = 3.1416$ [05]
- (c) Define Absolute error and relative error in numerical analysis. [02]

OR

Q.5. Answer the following questions. [14]

- (a) Determine the constants a and b by least-squares method such that $y = ae^{bx}$, [07]
fits the following data.

x	1.0	1.2	1.4	1.6
y	40.170	73.196	133.372	243.02

- (b) Find a real root of the equation $x^3 - 2x - 5 = 0$ using Bisection method in [05]
the interval $[2,3]$. Perform five iterations.
- (c) Explain False Position Method to find real approximate root of equation [02]
 $f(x) = 0$.

Q.6. Answer the following questions. [14]

- (a) Determine the value of y when $x = 0.1$ given that [04]
 $y(0) = 1$ and $y' = \frac{dy}{dx} = x^2 + y$ by using Euler's formula. Take $h = 0.05$.
- (b) Given $\frac{dy}{dx} = 1 + y^2$ where $y(0) = 0$, Find $y(0.2)$ and $y(0.4)$ correct to four [05]
decimals using the fourth order Runge-Kutta method. Use $h = 0.2$
- (c) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal places using trapezoidal [05]
rule with $h = 0.125$

OR

Q.6. Answer the following questions. [14]

- (a) From Taylor series for $y(x)$, find $y(0.1)$ correct to three decimal places if $y(x)$ [04]
satisfies $y' = \frac{dy}{dx} = -y$ and $y(0) = 1$
- (b) Solve the equation $\frac{dy}{dx} = 1 + xy$, up to second approximation using Picard's [05]
method where $y(0) = 0$.
- (c) Use Simpson 1/3 rule to evaluate approximate value of $\int_0^1 \frac{1}{1+x^2} dx$. Use $n = 4$ [05]
subintervals.